

# Daniel Andres Jimenez Riveros

Senior Data Scientist | NLP, LLMs & Generative AI

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## Professional Summary

**Senior Data Scientist / AI-ML Engineer with 5+ years of experience** applying data science to product development, with deep focus on Natural Language Processing (NLP), machine learning, and Large Language Model (LLM) systems. Master's degree in Computational Linguistics. Builds and operates production NLP and agentic LLM pipelines on AWS, and leads initiatives end to end from hypotheses and KPIs through deployment, monitoring, and measurable business impact.

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## Work Experience

### Apexon, Vancouver, BC, Canada

#### *Senior GenAI Engineer (Data Science, NLP & LLM)*

June 2025 - Present

- **Projected +\$0.8B net new revenue from prioritized ISV verticals within the first launch year** by leading an agentic AI solution for a payment platform (McKinsey client) from discovery through implementation, testing, deployment, and monitoring in Python, AWS, and LangGraph, applying agentic patterns such as planning, tool and function calling, multi-agent orchestration, and guardrails.
- **Achieved 96% accuracy in a production RAG system** by architecting NLP retrieval pipelines on AWS (Bedrock, OpenSearch, S3, Lambda) with Anthropic Claude and OpenAI LLM APIs and advanced prompt and context engineering; defined success metrics and ran evaluations with DeepEval and Ragas plus Langfuse observability for quality and drift monitoring.
- **Serve as technical lead and go-to resource for the data science team** on architecture, approach, and technical quality: established best practices for coding standards, peer review, reproducibility, testing, and CI/CD; coached junior members; and partnered with stakeholders, engineers, and business SMEs to translate business needs into scalable solutions, prioritizing and managing trade-offs across multiple parallel workstreams and communicating model and business KPIs to technical and non-technical audiences.

### Arkangel AI, Remote

#### *Senior Machine Learning Engineer (NLP / LLM)*

June 2024 - June 2025

- **45,000+ users engaged with an LLM agent for unstructured clinical text extraction** from electronic health records, built with Hugging Face Transformers, RAG, and LangGraph and operated as a reliable production NLP pipeline on AWS (Fargate, ECR, S3) with Docker and FastAPI; raised task accuracy through SFT, LoRA/PEFT fine-tuning, and RLHF/DPO alignment measured on a held-out evaluation set.
- **Generated up to \$50k monthly recurring revenue** by defining the hypotheses and KPI framework for, and delivering, an AutoML and deep learning platform on AWS SageMaker with PyTorch, Lambda, S3, and CloudWatch, including automated retraining, model monitoring, and A/B experimentation to validate impact before rollout.
- **Reduced processing time and cost by 70%** by delivering document AI with vision-language models, OCR, and computer vision (OpenCV, PyTorch) for automated invoice processing and provider verification, a claims-style document review workflow, de-risking an ambiguous problem space with staged prototyping and measurable success criteria.

### Amazon, Remote

#### *Software Development Engineer*

July 2022 - August 2023

- **Analyzed 500,000+ customer service transactions for 9,000+ Amazon employees** by building text analytics and automated data aggregation pipelines in Python (Pandas, NumPy) and full-stack analytical tools, recommending data-driven improvements from product and software usage patterns.
- **Empowered 300+ e-commerce leaders** by architecting scalable backend services and RESTful APIs in TypeScript and NodeJS with real-time data processing, peer code review, testing, and CI/CD.

#### *Information Technology Intern*

January 2021 - July 2021

- **Delivered analytics dashboards and internal tooling for 3 teams** using ReactJS and SAS-based analytics, turning stakeholder requirements into real-time visualizations.
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## Education

**Master of Data Science in Computational Linguistics**, University of British Columbia, Vancouver, BC, Canada (2023-2024)

**Master of Science in Information Technology**, Universidad de los Andes, Bogota, Colombia (2018-2020)

**Bachelor of Engineering in Industrial Engineering and Bachelor of Arts**, Universidad de los Andes, Bogota, Colombia (2014-2018)

**Graduate Diploma in Project Management**, Fundacion Universitaria Jorge Tadeo Lozano, Bogota, Colombia (2017)

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# Selected Projects

- Thorcast AI, Risk Modelling Platform for Financial Services

2025 - Present

  - Generating \$5k+ monthly revenue from banks with an on-premise application that builds credit and operational risk ML/DL models in minutes instead of months, automating data ingestion, feature engineering, model training, and evaluation with an AI agent in Python, FastAPI, LangGraph, PyTorch, and Ollama for on-premise LLM inference under strict data-privacy and regulatory constraints.
- Alzheimer's Detection from Speech, UBC Master's Thesis

March 2024 - August 2024

  - Achieved 90% recall by fine-tuning transformer-based NLP models and CNNs in PyTorch and TensorFlow for multi-modal analysis of speech transcripts and imaging, with hyperparameter tuning, transfer learning, and rigorous statistical evaluation.

## Awards & Achievements

COLFUTURO Scholarship (2023) | Data Science for All, DS4A (2018)

## Skills

**NLP & Large Language Models:** Natural Language Processing, Generative AI (GenAI), LLM API Integration (Anthropic Claude, OpenAI, Amazon Bedrock), Prompt Engineering, Context Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Vector Databases (OpenSearch, Pinecone, ChromaDB, FAISS), Fine-Tuning (SFT, LoRA, PEFT), RLHF/DPO Alignment, LLM Evaluation (DeepEval, Ragas), Guardrails, Text Classification, Named Entity Recognition, Information Extraction

**Agentic AI:** LangGraph, LangChain, Agent Planning, Tool and Function Calling, Multi-Agent Orchestration, Model Context Protocol (MCP), Human-in-the-Loop, Agent Observability (Langfuse)

**Machine Learning & Deep Learning:** PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, Supervised & Unsupervised Learning, Statistical Modelling, Computer Vision (OpenCV, VLMs, OCR), Feature Engineering, Model Evaluation, Hyperparameter Tuning

**Programming & Engineering:** Python (Expert), SQL, JavaScript, TypeScript, C++, Java, C#/.NET, Git & GitHub, CI/CD, Unit & Integration Testing, Code Review, Reproducibility, RESTful APIs, FastAPI, Microservices

**AWS & MLOps:** AWS (SageMaker, Bedrock, Lambda, S3, OpenSearch, Fargate, ECS, CloudWatch), Azure, Docker, Kubernetes, MLflow, Weights & Biases, Terraform/IaC, Model Deployment, Model Monitoring & Drift Detection, Production ML Pipelines

**Product & Analytics:** KPI Frameworks (OKRs, North Star Metrics, Model & Business KPIs), Hypothesis Definition, A/B Testing & Experimentation, Data Analysis (Pandas, NumPy), Stakeholder Management, Cross-Functional Collaboration, Technical Leadership, Mentoring & Coaching